

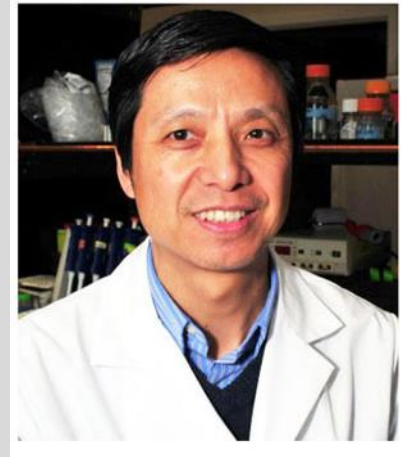
Foundations of Osteopathic Medicine

Signal Transduction II: Enzyme-linked Receptors and Nuclear Receptors

Qiangrong Liang, MD/PhD
Department of Biomedical Sciences
Lab 027, Office 028, Riland Building
Office Phone: 516-686-1331
Email: qliang03@nyit.edu

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Do.
Make.
Heal.

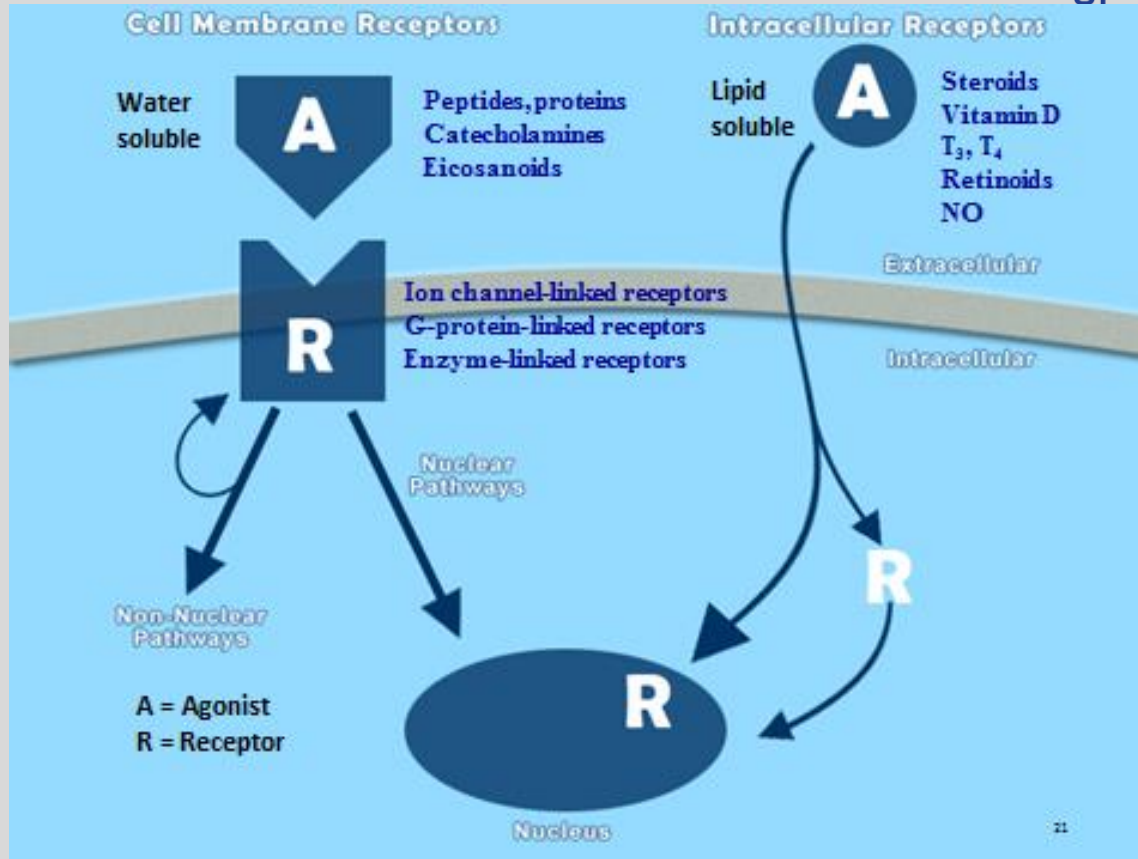
Innovate.
Reinvent the Future.

Session Objectives

1. **Compare and contrast** how atrial natriuretic peptide (ANP) and nitric oxide (NO) activate the two major types of guanylyl cyclase.
2. **Define** common enzyme-linked receptors and **outline** the mechanism of receptor tyrosine kinases (RTKs), providing examples of hormones or agonists that act through these receptors.
3. **Explain** the role of small G-proteins, such as Ras, in intracellular signaling.
4. **Outline** the MAP kinase and PI3K–Akt–mTOR signaling pathways.
5. **Describe and explain** the mechanism of action of intracellular (“nuclear”) receptors.

Types of Receptors

ge of Osteopathic
cine

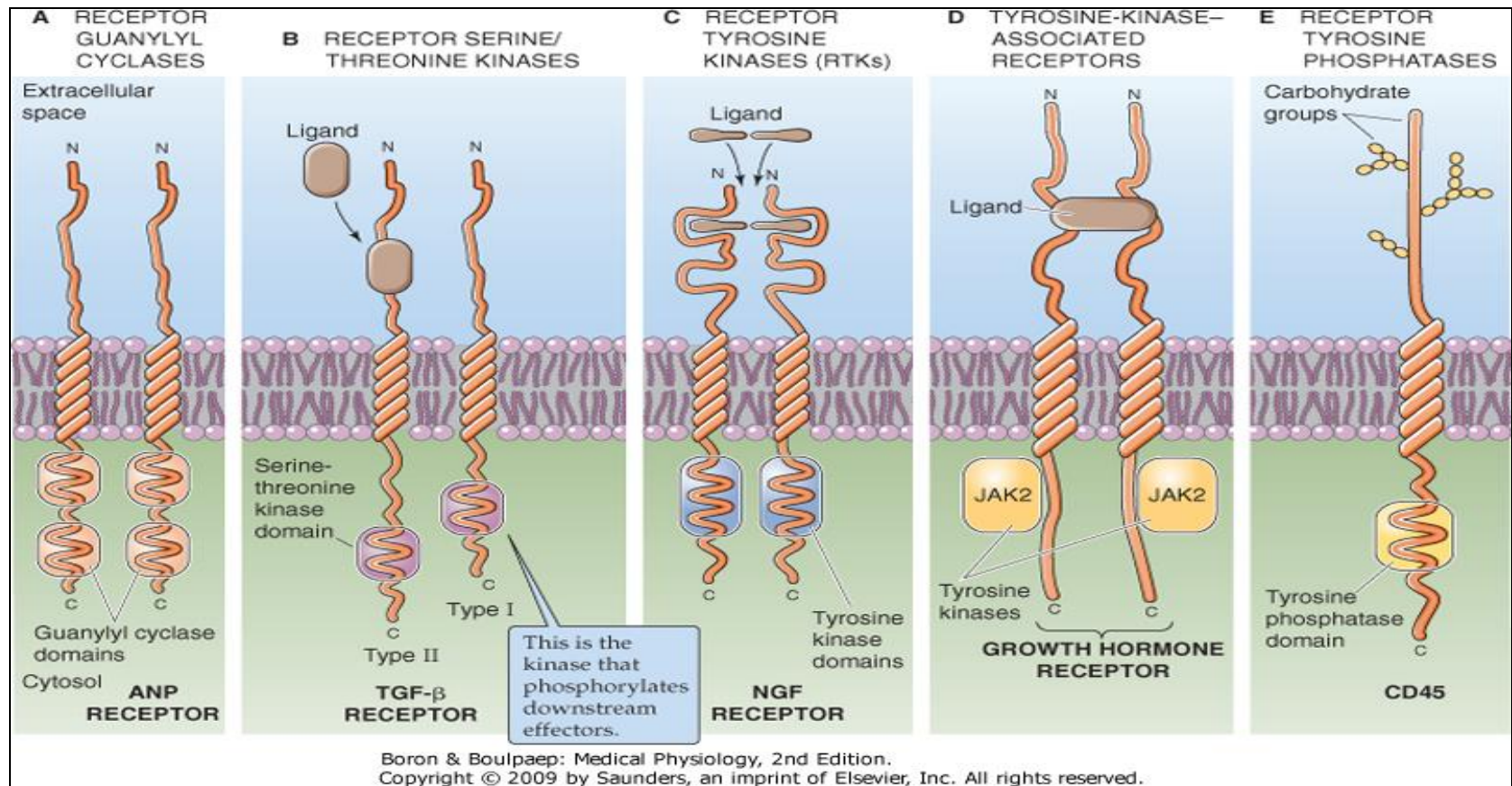


Enzyme-linked Receptors

- An Enzyme-linked receptor is not associated with a G protein. Instead, it has a latent enzyme activity that is activated when the receptor's specific agonist binds.
- Enzyme-linked receptors respond to numerous extracellular signals including various growth factors and other extracellular signaling molecules.
- The responses to these receptors are usually slow since they require many intracellular steps that lead to changes in gene expression.

Types of Enzyme-linked Receptor

- Receptor guanylyl cyclases
- Receptor tyrosine kinases
 - Growth, proliferation, differentiation, survival, and cell motility factors
- Receptor serine/threonine kinases
 - Transforming growth factor β (TGF- β) superfamily – development
- Tyrosine-kinase-*associated* receptors
 - e.g., “Cytokine receptors”
- Receptor tyrosine phosphatases

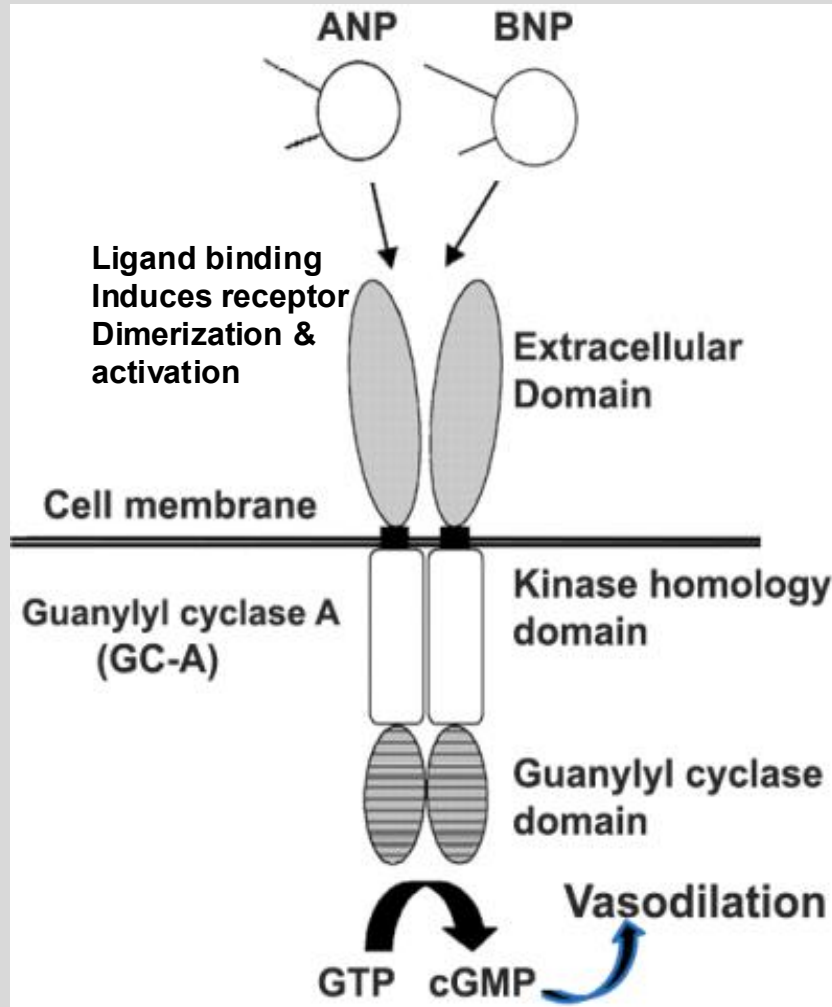


Catalytic receptors. A, Receptor guanylyl cyclases have an extracellular ligand-binding domain. B, Receptor serine/threonine kinases have two subunits. The ligand binds only to the type II subunit. C, Receptor tyrosine kinases (RTKs) similar to the NGF receptor dimerize on binding a ligand. D, Tyrosine kinase-associated receptors have no intrinsic enzyme activity but associate noncovalently with soluble, nonreceptor tyrosine kinases. E, Receptor tyrosine phosphatases have intrinsic tyrosine phosphatase activity. ANP, atrial natriuretic peptide; JAK, Janus kinase (originally "just another kinase"); NGF, nerve growth factor; TGF- β , transforming growth factor β .

Receptor Guanylyl Cyclases

Two types of receptor guanylyl cyclase:

- The receptor for atrial natriuretic peptide (ANP) and B-type natriuretic peptide (BNP) is a membrane-bound guanylyl cyclase A (**GC-A**). ANP and BNP are degraded by **Neprilysin**, a neutral endopeptidase.
- The other form of receptor GC is a cytoplasmic soluble GC (**sGC**) that is the target of nitric oxide (NO), a membrane-soluble gas.
- Both GC-A and sGC produce cGMP from GTP
- There are drugs that target **GC-A** and **sGC** pathways

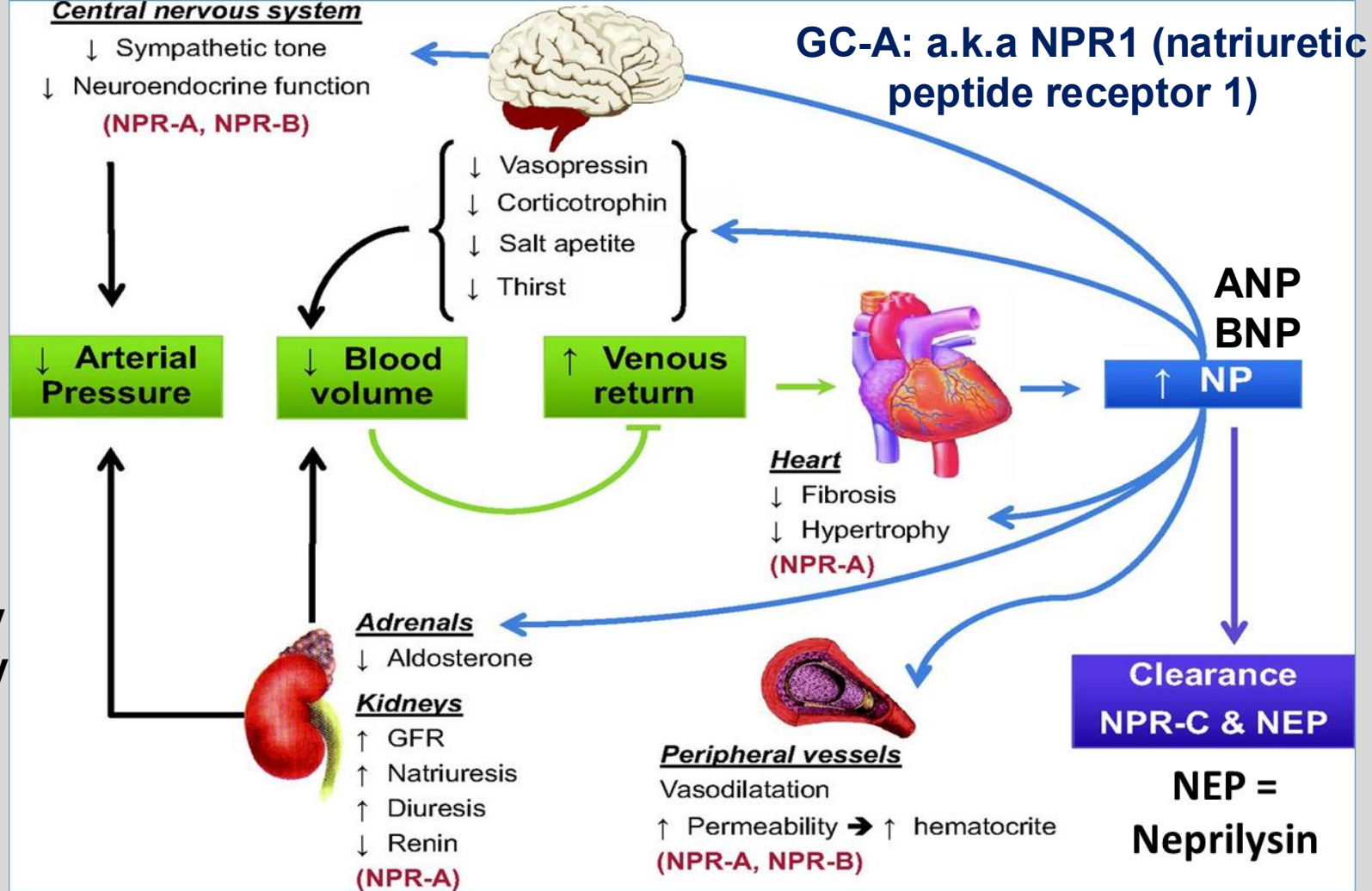


**A membrane-bound
receptor with GC activity**

**Structure of the ANP/BNP
Receptor: GC-A**

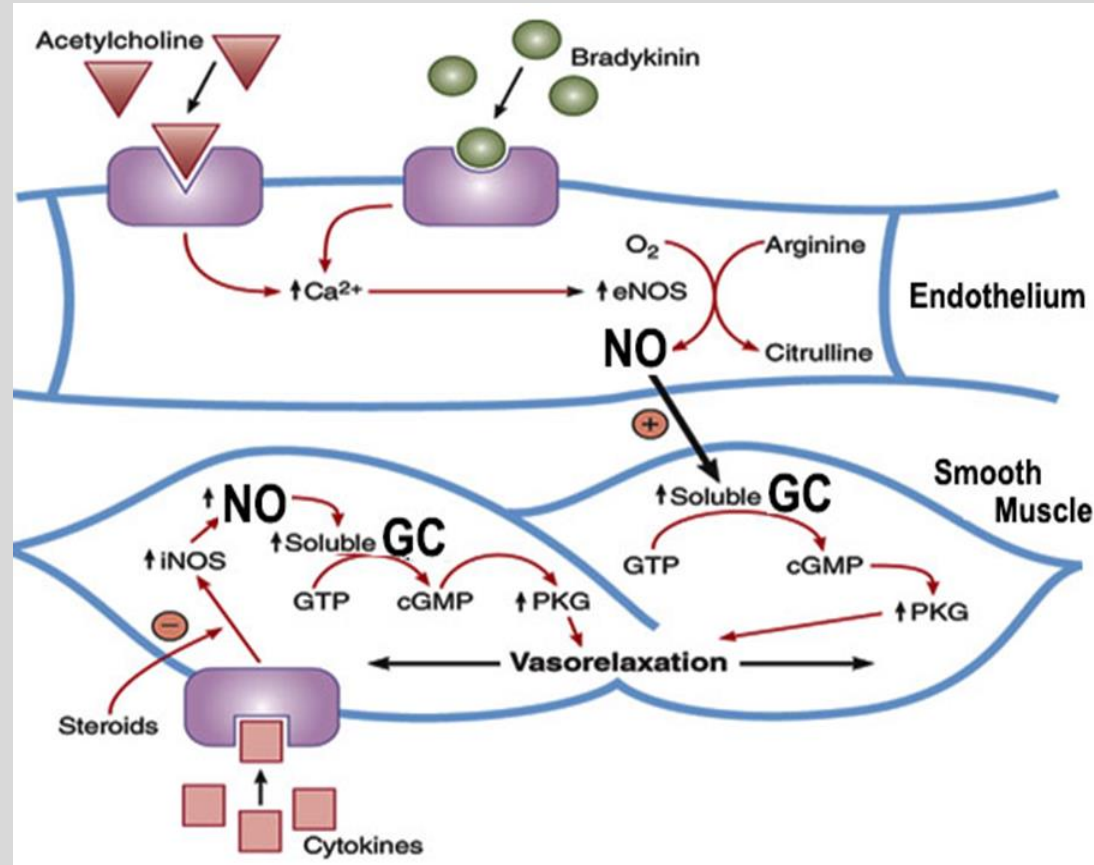
GC-A: Function

Heart and
Circulatory
Physiology
2013; Vol.
304:H358



Nitric oxide (NO)- guanylate cyclase (sGC)

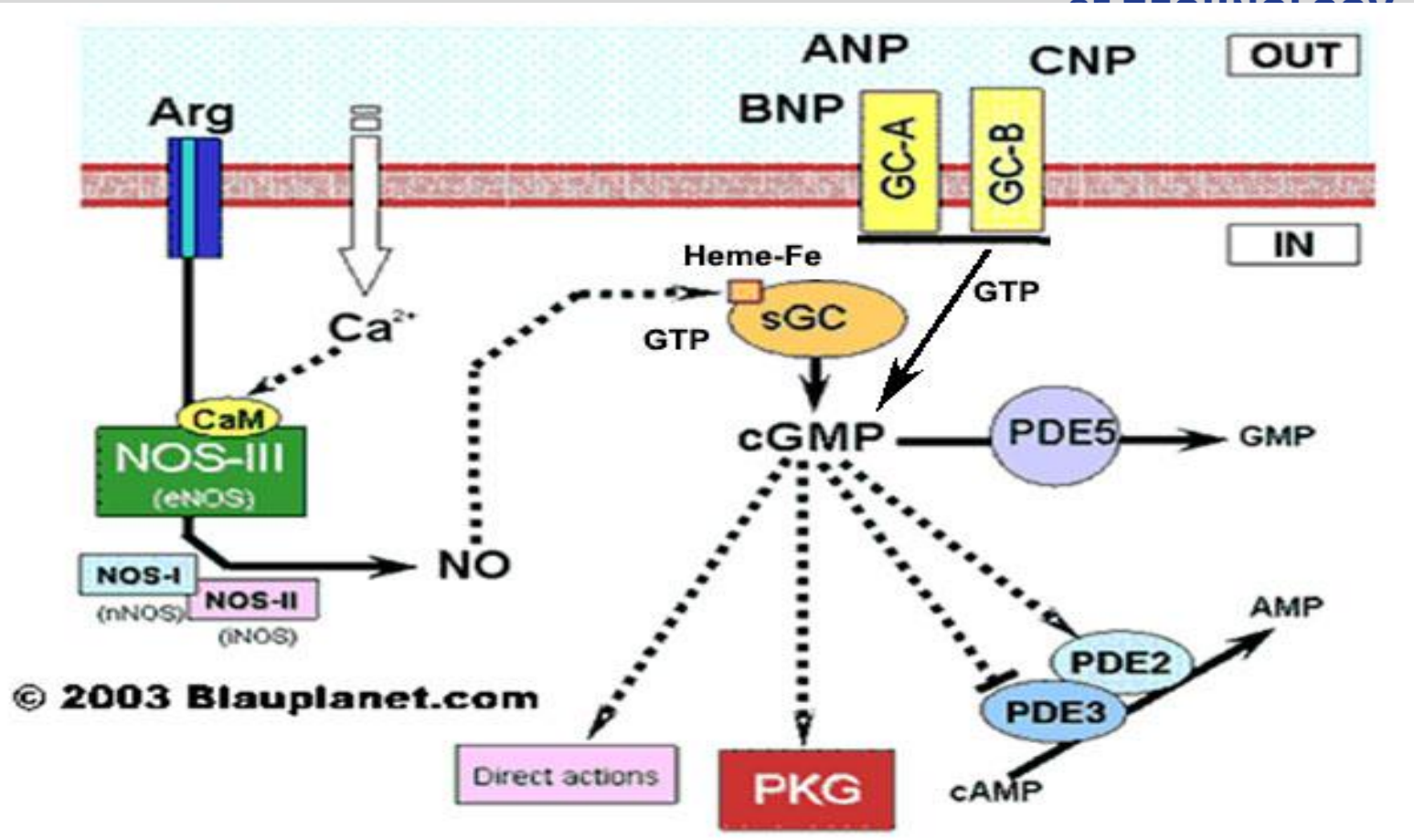
- NO is a *gas* signaling molecule essential for the regulation of blood flow and blood pressure
- NO is produced in vascular endothelial cells by the enzyme nitric oxide synthase (NOS).
- NO diffuses to nearby smooth muscle cells, where it causes vasodilation
- NO does so by activating its soluble receptor guanylate cyclase (sGC) which produces cGMP.



Citation: Hormones and Hormone Action, Gardner DG, Shoback D. *Greenspan's Basic & Clinical Endocrinology*, 10e; 2017. Available at: <https://accessmedicine.mhmedical.com/content.aspx?bookid=2178§ionid=166246760>
Accessed: January 15, 2020

Copyright © 2020 McGraw-Hill Education. All rights reserved

Both ANP/BNP and NO stimulate cGMP Production



Drugs targeting NO-cGMP

- **Nitroglycerin**: Breaks down spontaneously to release NO. This causes (1) relaxation of cardiac arterioles, which improves cardiac perfusion, and (2) relaxation of peripheral arterioles, which reduces the work of the heart, both of which relieve the pain of *angina pectoris*.
- **Sildenafil (Viagra)**: Elevates cGMP levels by inhibiting phosphodiesterase type 5 (PDE5), an enzyme that promotes degradation of cGMP, thereby increasing blood flow to the penis. Sildenafil is also used to treat pulmonary arterial hypertension.

Receptor Tyrosine kinases

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- **Why should we care about Receptor Tyrosine Kinases (RTKs)?** **Insulin** and many **growth factors** exert their effect through RTKs: diabetes, tumors, etc.
- **A kinase** is an enzyme that transfers phosphate from ATP to specific target molecules; this process is an important mechanism for signal transduction.
- **Tyrosine kinases** are enzymes that phosphorylate tyrosine amino acids.
- **RTKs** are transmembrane proteins that act as receptors and have intrinsic tyrosine kinase activities. Humans have 58 genes for 20 families of receptor tyrosine kinases, each with distinct structural features.
- **Structurally**, RTKs each have an extracellular ligand binding domain connected to a cytoplasmic tyrosine kinase domain by a single transmembrane helix with a carboxyl terminal segment having multiple tyrosines for phosphorylation

Seven subfamilies of receptor tyrosine kinases

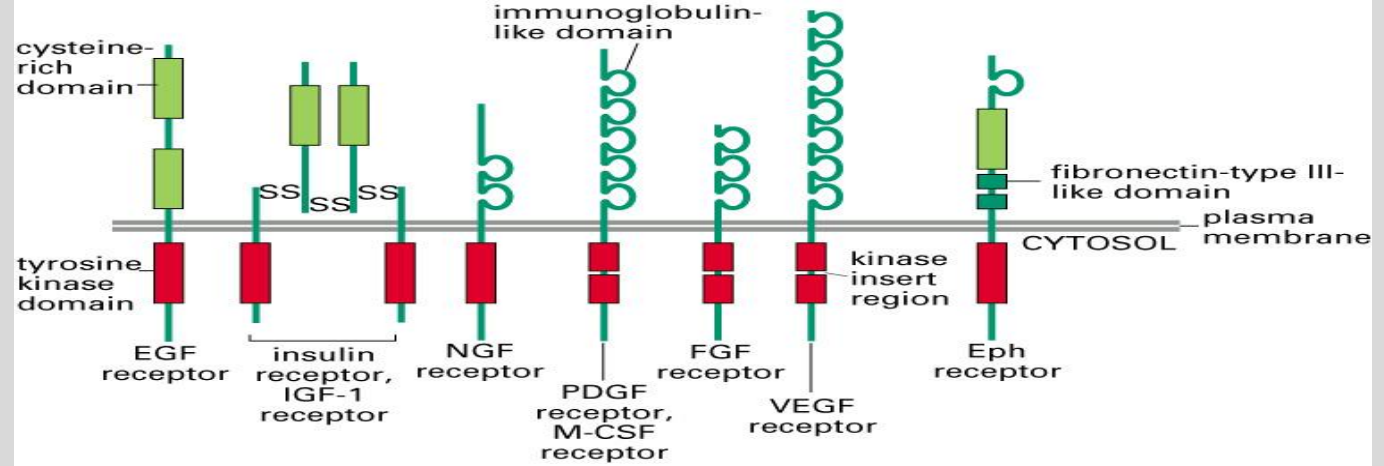
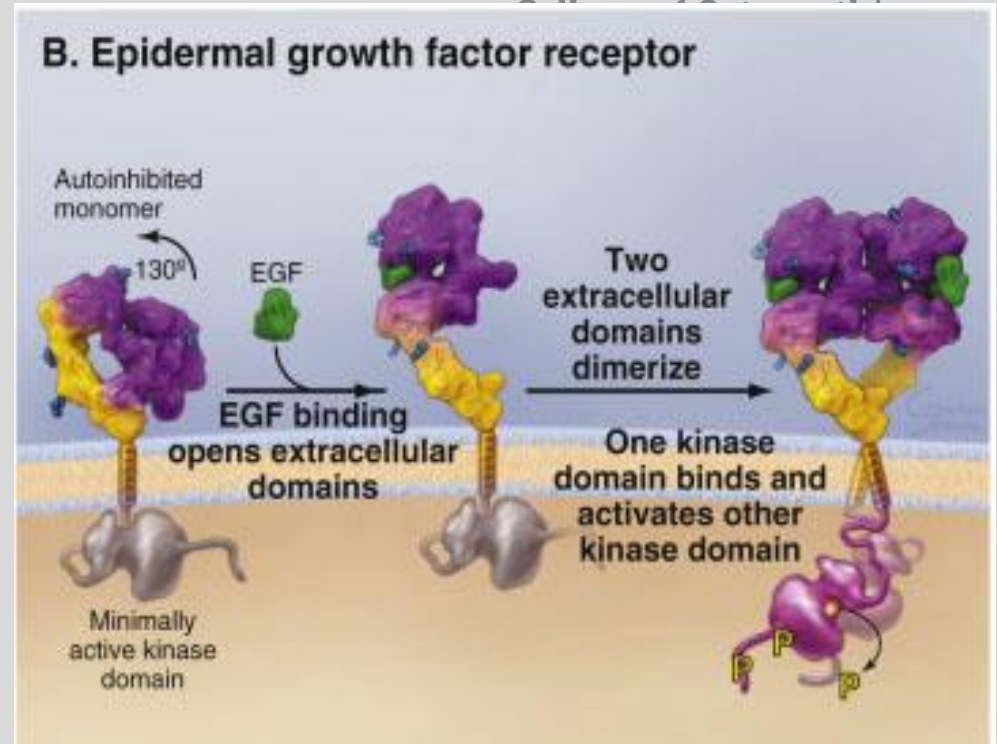


TABLE 15-4 Some Signaling Proteins That Act Via Receptor Tyrosine Kinases

SIGNALING LIGAND	RECEPTORS	SOME RESPONSES
Epidermal growth factor (EGF)	EGF receptor	stimulates proliferation of various cell types
Insulin	insulin receptor	stimulates carbohydrate utilization and protein synthesis
Insulin-like growth factors (IGF-1 and IGF-2)	IGF receptor-1	stimulate cell growth and survival
Nerve growth factor (NGF)	Trk A	stimulates survival and growth of some neurons
Platelet-derived growth factors (PDGF AA, BB, AB)	PDGF receptors (α and β)	stimulate survival, growth, and proliferation of various cell types
Macrophage-colony-stimulating factor (M-CSF)	M-CSF receptor	stimulates monocyte/macrophage proliferation and differentiation
Fibroblast growth factors (FGF-1 to FGF-24)	FGF receptors (FGF-R1–FGF-R4, plus multiple isoforms of each)	stimulate proliferation of various cell types; inhibit differentiation of some precursor cells; inductive signals in development
Vascular endothelial growth factor (VEGF)	VEGF receptor	stimulates angiogenesis
Ephrins (A and B types)	Eph receptors (A and B types)	stimulate angiogenesis; guide cell and axon migration

Receptor Tyrosine kinases, cont.

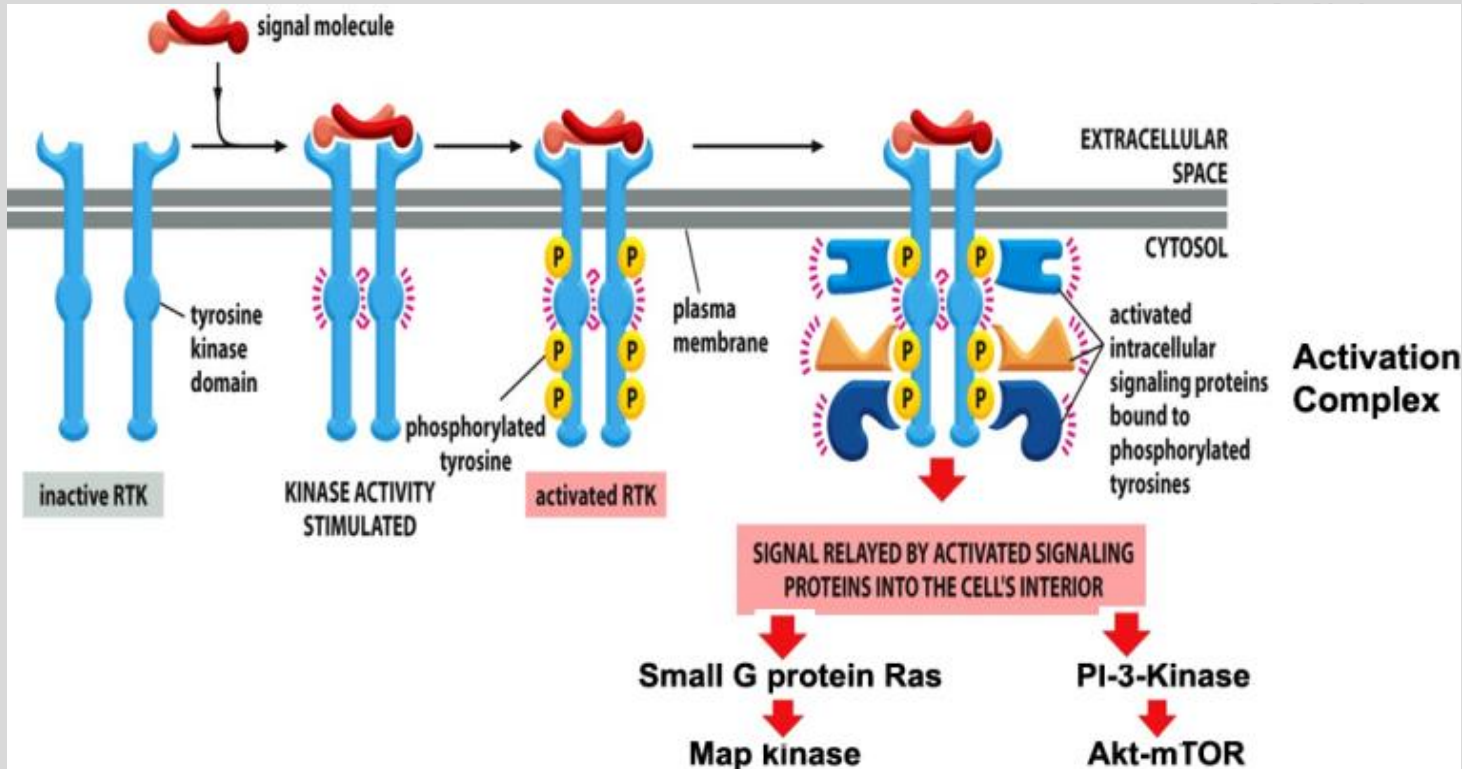
- Typically, agonist binding to the extracellular domain of a RTK causes the receptor molecules to form a dimer or multimer which activates their intrinsic tyrosine kinase activity.
- The activated kinase phosphorylates certain tyrosine residues of the receptor itself which then assembles a complex of intracellular signaling proteins



RTK and Integration Center

- RTK activation stimulates the assembly of an intracellular signaling complex which:
 1. Relay the initial “message” into the cell interior and cross-talk with other pathways.
 2. Also receive and integrate signals from a variety of intracellular components, including those from other transduction pathways such as GPCRs.
 3. Coordinate and regulate complex cell activities such as survival, growth, differentiation, proliferation, and motility.
 4. It serves as an integration center of different signals, leading to the activation of many downstream pathways.

Most RTK can activate two signal transduction pathways: the Ras-MAP kinase and the PI-3-kinase-AKT-mTOR pathway



Small (monomeric) G proteins

- Many tyrosine kinase receptors activate a second type of G protein, known as small or monomeric G proteins.
- Small G proteins are like the large heterotrimeric G proteins
 - They bind GTP, absolutely require it for activity, and are inactive when bound to GDP.
 - They have an innate GTPase activity to turn themselves off and serve as molecular switches in the cell.
 - They act as intermediaries between their receptor and more distal parts of the cascade.
- However, small G proteins are structurally distinct from the heterotrimeric G proteins and arise from different genes. Three or more classes of the small G proteins are known. We'll discuss **Ras family** only.

Ras

- Ras is a superfamily of small G proteins.
- It regulates organ development and growth as well as cell proliferation, survival and death.
- Ras protein is activated by most growth factor-activated receptor tyrosine kinases.
- When activated, the Ras protein sets in motion a cascade to activate the mitogen-activated protein (MAP) kinase module.
- Mutated Ras genes found in 30% human cancers.
- The mutated Ras genes are called oncogenes, while the normal genes proto-oncogenes.
- Mutant Ras can promote abnormal cell growth leading to cancer formation.

The activation of Ras-MAP Kinase by RTK signaling

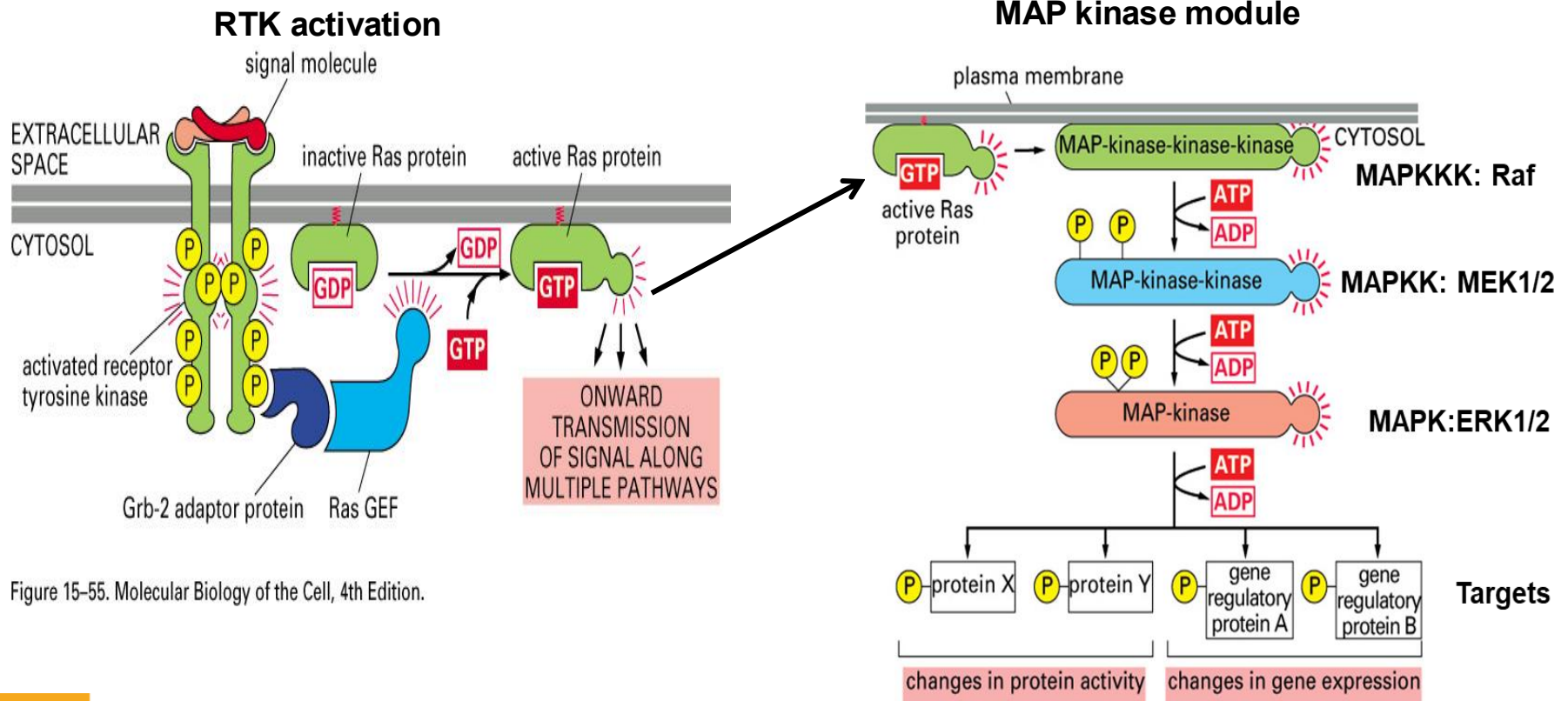


Figure 15-55. Molecular Biology of the Cell, 4th Edition.

RTKs activate the PI-3-kinase-AKT-mTOR pathway

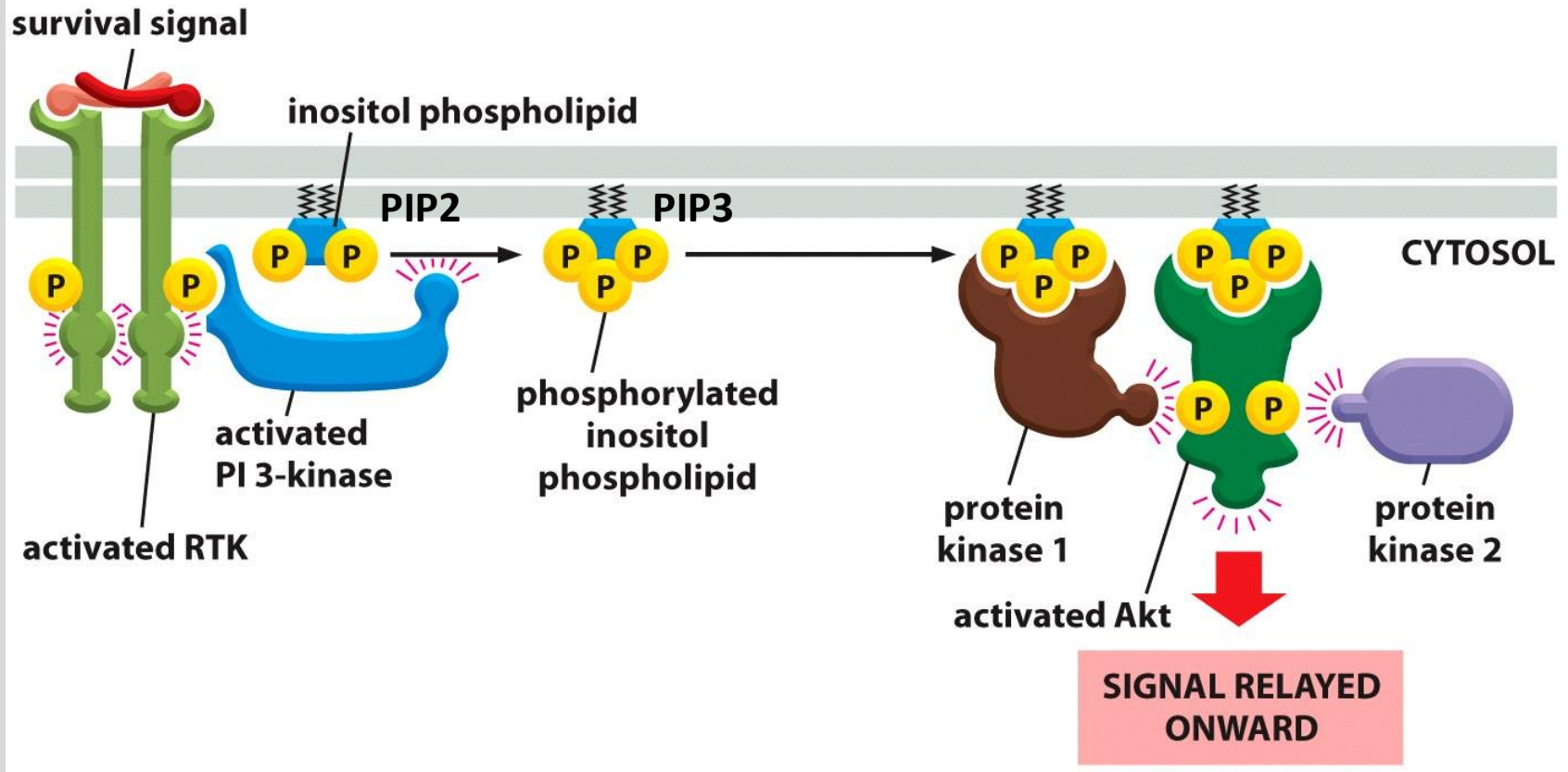
NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

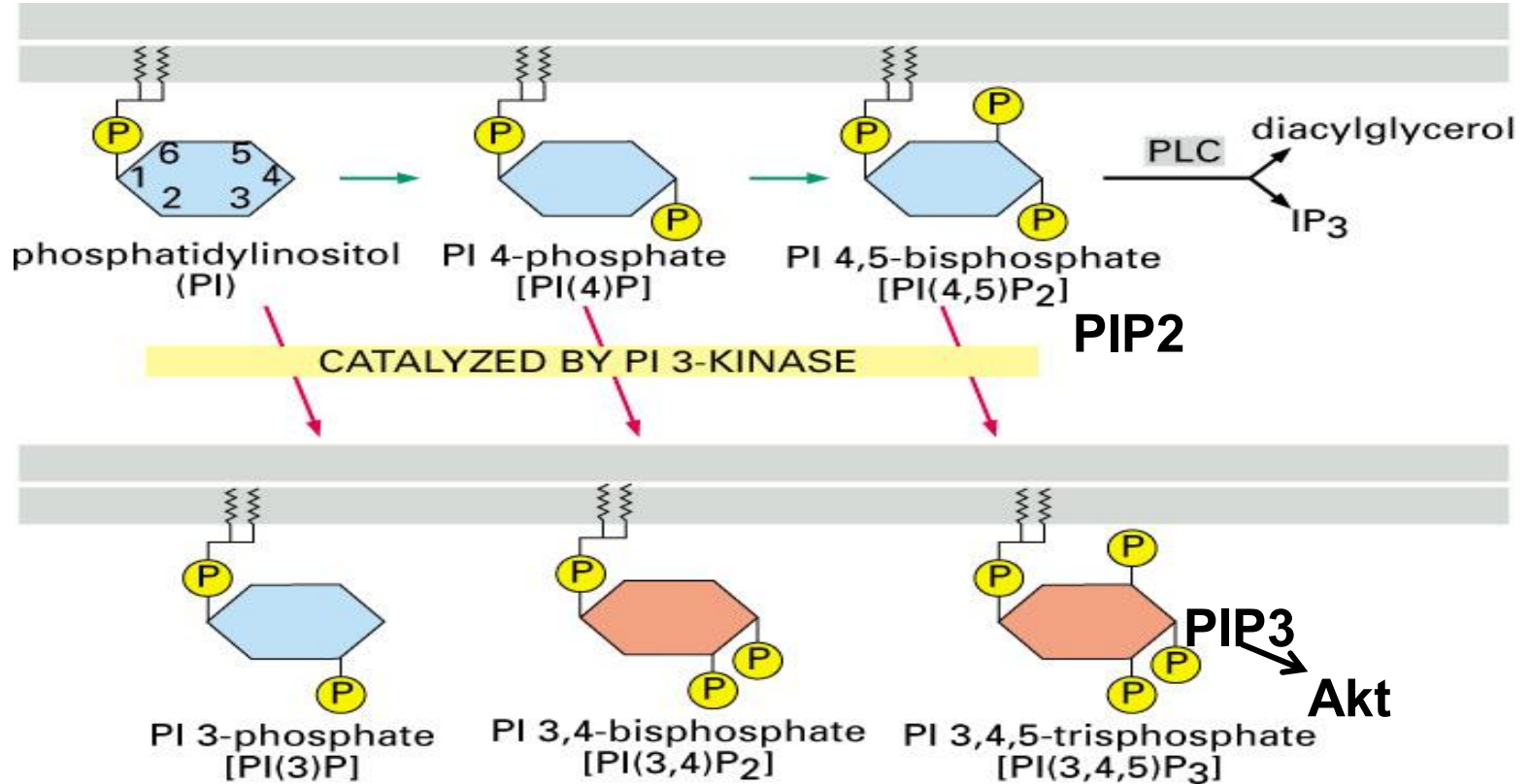
- Most RTKs that promote cell growth and survival activate the enzyme phosphatidylinositide 3-kinase (PI-3-kinase).
- PI-3-kinase phosphorylates inositol phospholipids in the plasma membrane which serve as docking and activation sites for several intracellular proteins including Akt.
- Activated Akt phosphorylates and inactivates a pro-apoptotic protein Bad to promote cell survival.
- Akt is required for the metabolic responses to insulin, including Glut 4 translocation to cell membrane to enhance glucose uptake
- Akt also helps activate the mechanistic target of rapamycin (mTOR), a serine/threonine kinase critical for cell growth.
- PI-3-kinase-Akt-mTOR pathway is an important drug target for cancer, heart disease and other conditions.

RTK activates the PI-3-kinase-AKT signaling pathway

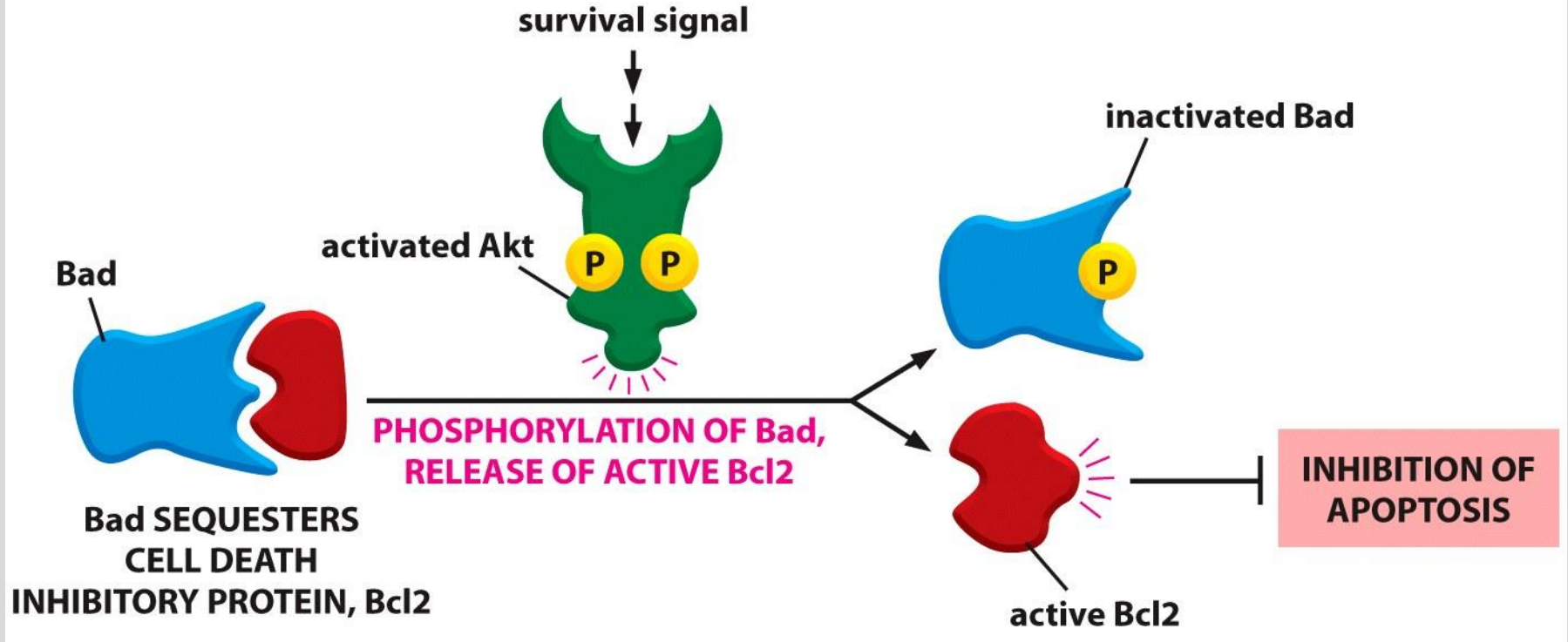
NEW YORK INSTITUTE
OF TECHNOLOGY

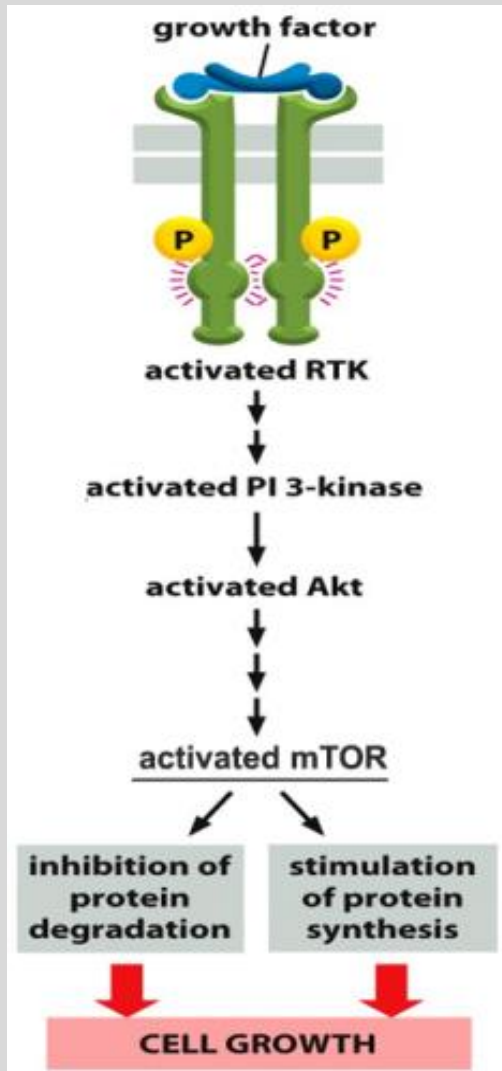


The inositol phospholipids generated by PI3K



Activated Akt promotes cell survival by inactivating Bad



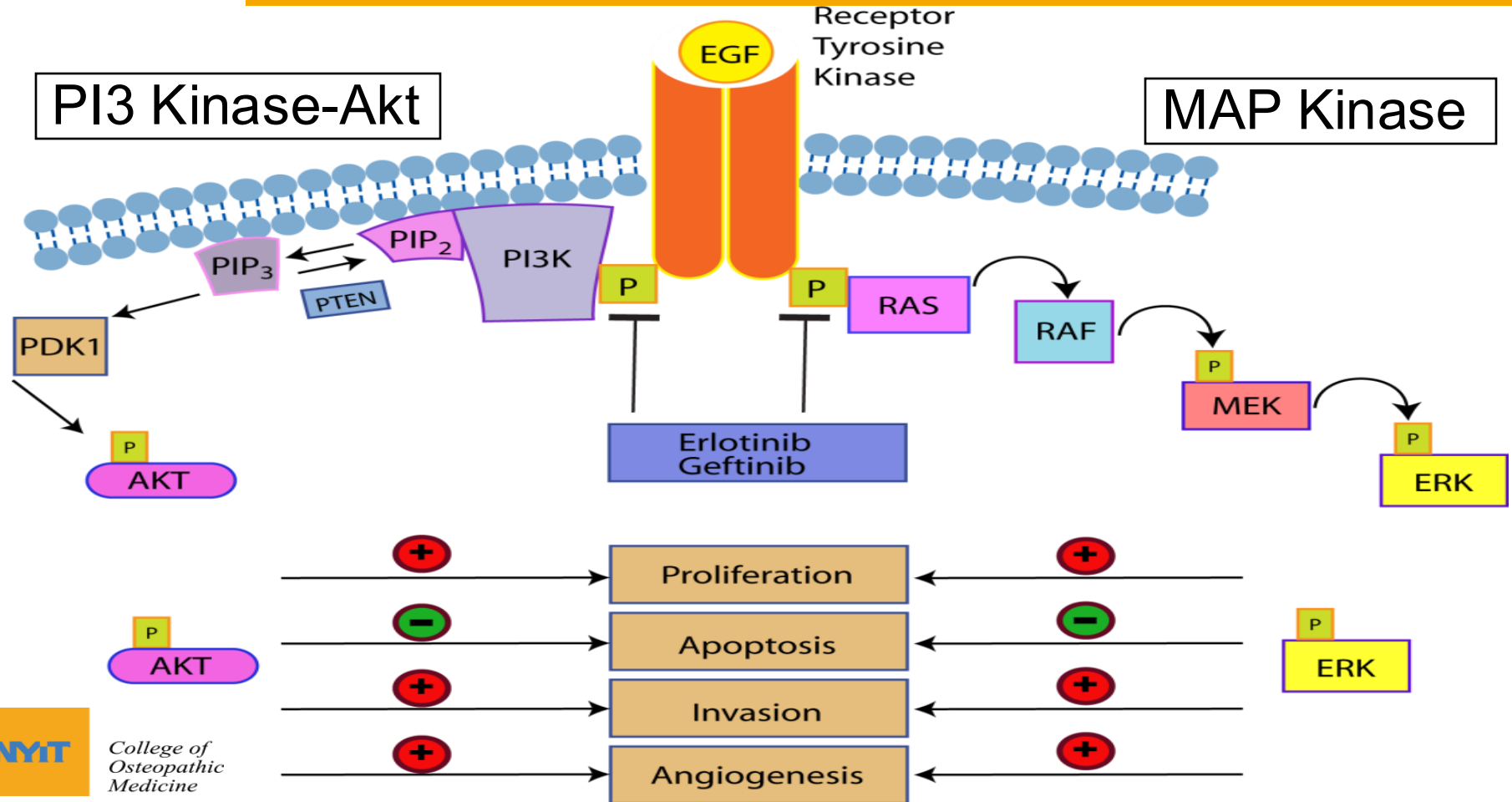


NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Akt promotes cell growth through the mechanistic target of rapamycin (mTOR), a serine/threonine protein kinase

Targeting RTK, AKT and ERK for Tumor Chemotherapy



Tyrosine kinase-associated receptors

- Receptors for some growth factors (GF) and “cytokines” are not themselves enzymes but can activate **loosely-associated tyrosine kinases** such as **JAK** (Janus kinase).
- This family includes receptors for GH, prolactin, leptin, interleukins, interferons, erythropoietin, and granulocyte-macrophage colony-stimulating factor (GM-CSF), and more.
- GH or cytokines induces receptor dimerization and activates JAK which then phosphorylates the receptor and the STAT proteins (signal transducers and activators of transcription). The STATs then associate with other STAT members to form a complex that modulates gene expression in the nucleus.

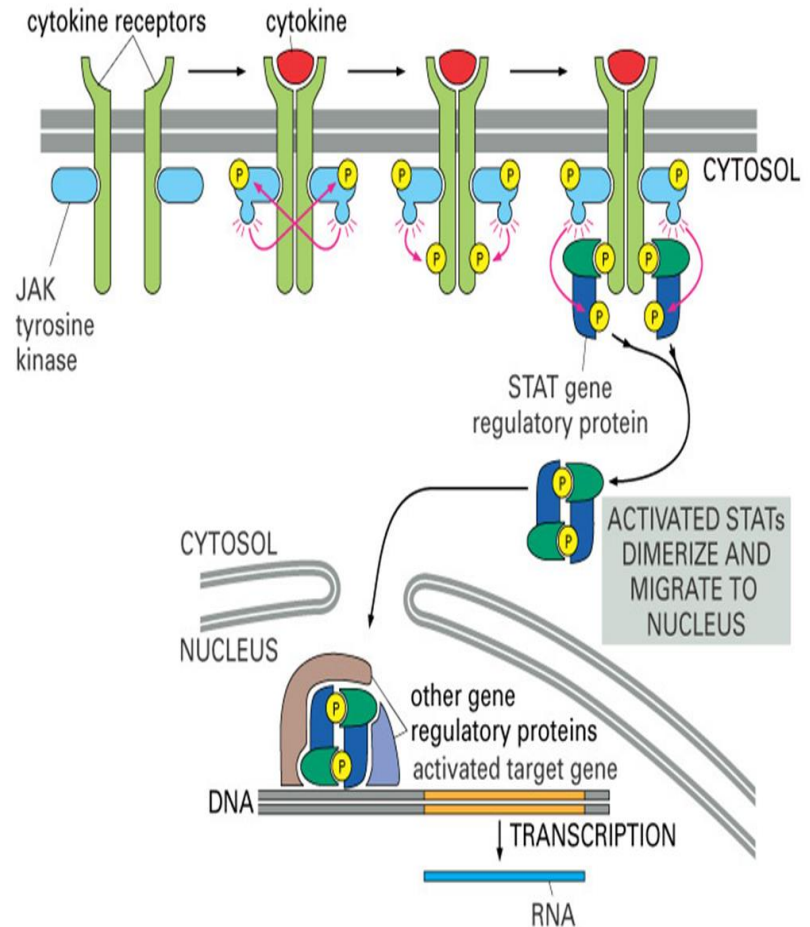


Figure 16-36 Essential Cell Biology, 2/e. (© 2004 Garland Science)

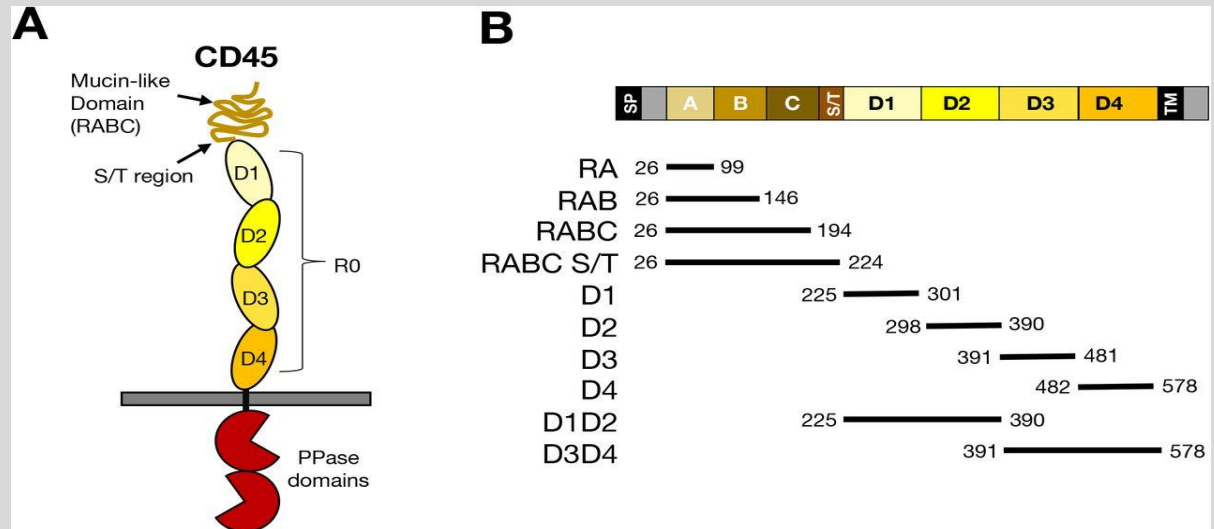
Receptor Tyrosine phosphatases

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

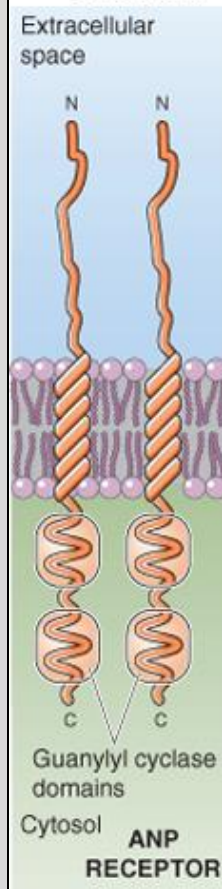
- **Phosphatases** remove phosphate groups from proteins, opposing kinase activity.
- **Receptor protein tyrosine phosphatases (RPTPs)** remove phosphates from tyrosine residues, counteracting RTK signaling.
- Many RPTPs are enriched in immune cells and regulate immune activation/suppression.
- **Example: CD45 –**

CD45 expressed on all leukocytes; regulates lymphocyte signaling by controlling **Src family kinase** phosphorylation.

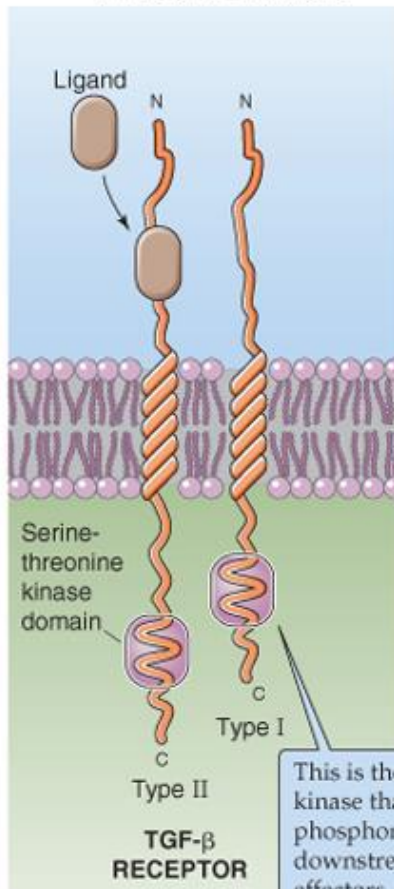


Human CD45, including the extracellular mucin and fibronectin like domains and intracellular phosphatase domains.

A RECEPTOR
GUANYLYL
CYCLASES

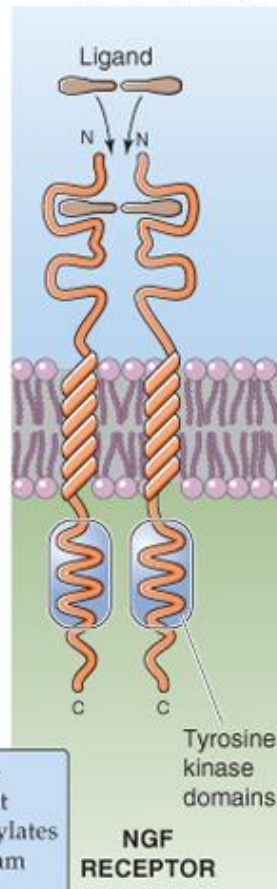


B RECEPTOR SERINE/
THREONINE KINASES

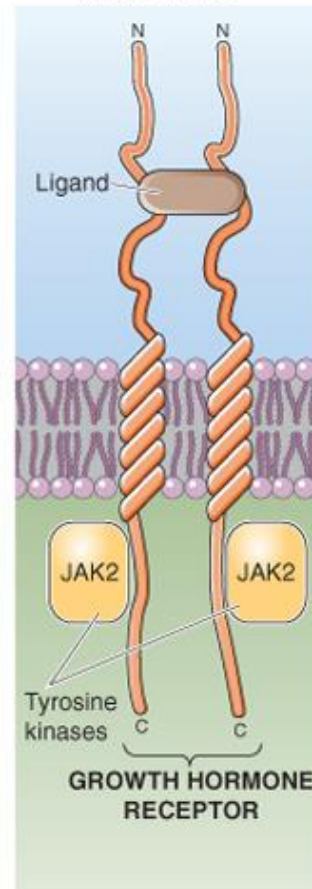


This is the kinase that phosphorylates downstream effectors.

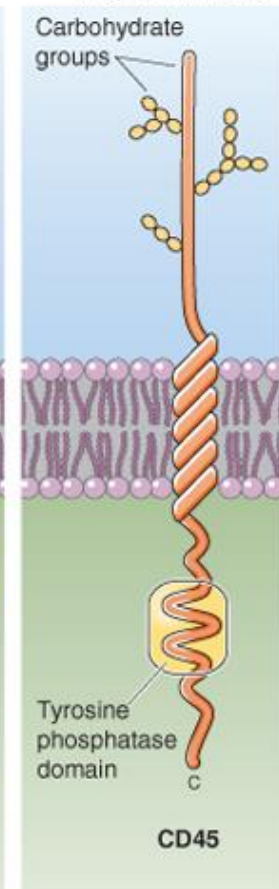
C RECEPTOR
TYROSINE
KINASES (RTKs)



D TYROSINE-KINASE-
ASSOCIATED
RECEPTORS

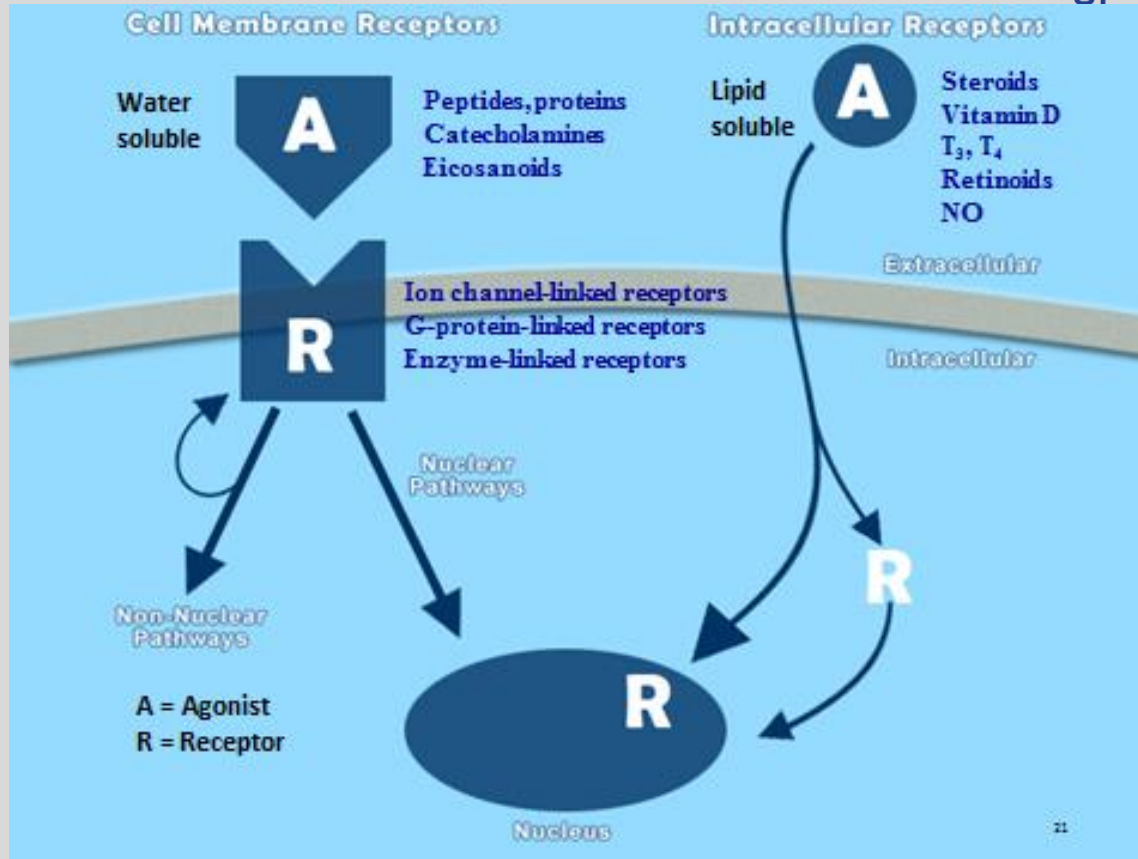


E RECEPTOR
TYROSINE
PHOSPHATASES



Types of Receptors

ge of Osteopathic
cine



Intracellular receptors a.k.s Nuclear receptor

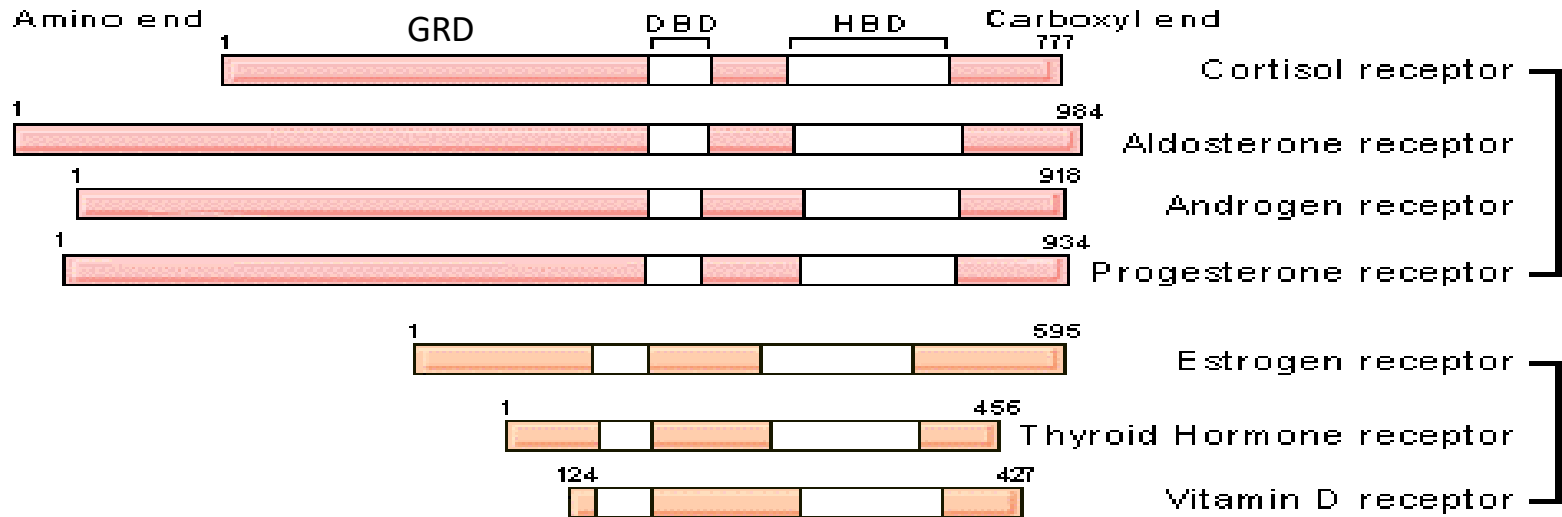
- They are also called nuclear receptors, although some are located initially in the cytoplasm. The similar receptors may be also localized to plasma membrane.
- Whatever the initial location of the receptor, the ligand-receptor complex translocates to the nucleus (if not already there) to perform its function there.
- Intracellular/nuclear receptors are **transcription factors** which are activated by their agonists and regulate gene transcription.

Structure of intracellular receptors

HBD: hormone-binding domain

DBD: DNA binding domain

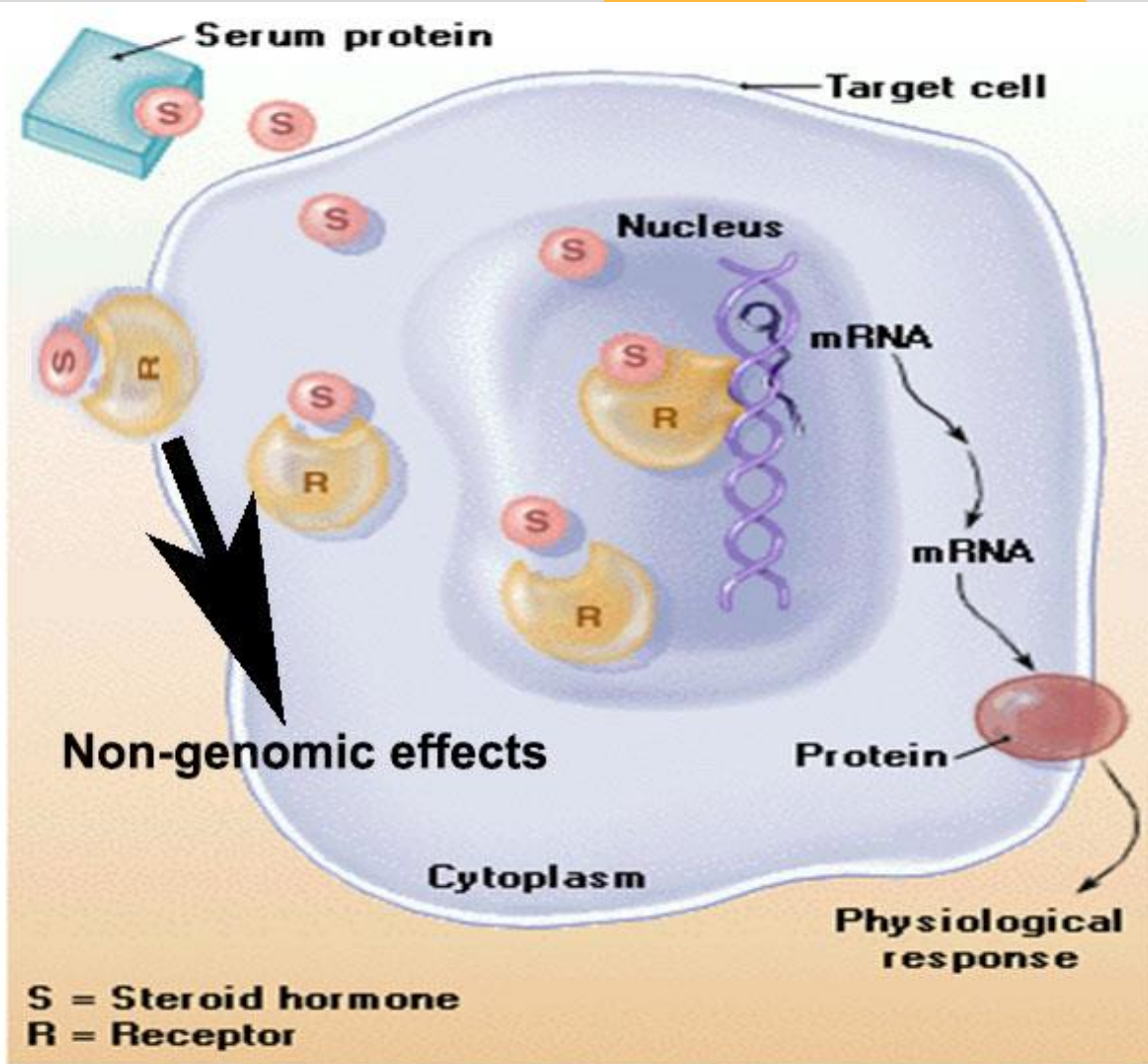
GRD: Gene regulatory domain



Hormones that Bind to Intracellular Receptors

- Such hormones are lipid-soluble: steroids (derivatives of cholesterol), thyroid hormones (T_3 , T_4), vitamin A and D, and Retinoids.
- These hormones bind and activate their respective receptors which then translocate to the nucleus and bind to DNA to affect the expression of their target genes. It can take minutes, hours, or days for the hormone to induce a biological response in the target cell.
- However, these hormones can also produce rapid cellular effects (seconds or minutes) which do not depend on gene transcription or protein synthesis. These are called **non-genomic effects** which are mediated through membrane receptors (steroids, $T_3/4$ and vitamin D).
- Membrane receptors have the same protein structure as intracellular receptors but are instead localized to membrane caveolae.
- Both genomic and non-genomic effects are believed to be important for normal function of these hormones.

Genomic and non-genomic effects of Steroid hormones



Summary of the Session

1. ANP and NO activate membrane-bound guanylyl cyclase and soluble guanylyl cyclase respectively to produce cGMP.
2. Insulin, IGF-1, PDGF, FGF and EGF activate receptor tyrosine kinases which phosphorylate the tyrosine residues on the receptor leading to the formation of activation complexes and the activation of intracellular pathways such as Ras.
3. Small G protein Ras work through kinase cascades such as MAP kinase to regulate cell proliferation, survival and death.
4. PI-3-Kinase-Akt-mTOR pathway is activated by RTK and plays important role in cell growth and survival.
5. Nuclear receptors mainly bind to DNA to regulate gene expression but can also induce quick non-genomic effects.

Question 1

A new experimental drug is being tested for the treatment of melanoma. In cultured tumor cells, the drug prevents ligand-induced activation of a membrane receptor, thereby blocking downstream activation of Ras and the MAP kinase cascade. Which of the following combinations of signaling components normally occur together in this type of pathway?

- a. Insulin; nuclear receptor; inositol trisphosphate; nitric oxide
- b. Receptor dimerization; auto-phosphorylation; Ras; MAP-kinase
- c. G-protein coupled receptors; auto-phosphorylation; MAP-kinase
- d. Steroid hormones; phospholipase A2; inositol trisphosphate; nitric oxide
- e. Vitamin D; a 7-pass receptor; phospholipase A2; nitric oxide

Question 2

A 55-year-old man is undergoing surgical resection of a colonic adenocarcinoma. Molecular analysis of the tumor shows a mutation in a GTP-binding protein that is constitutively active due to loss of its intrinsic GTPase activity. This protein functions downstream of receptor tyrosine kinases and activates the MAP kinase signaling cascade. Which of the following statements is true regarding this protein?

- a. Mutant forms are present in many human cancers.
- b. Once activated, it remains active until degraded to their constituent amino acids.
- c. It is a kinase that activates phospholipases.
- d. It always regulates the activity of G protein coupled receptors.
- e. They control the production of nitric oxide.

Question 3

A 52-year-old woman presents with a painless lump in her left breast. A biopsy reveals malignant epithelial cells.

Genetic testing shows the tumor cells harbor a point mutation in Ras that locks the protein in its GTP-bound active form, preventing GTP hydrolysis. Which of the following best describes the downstream effect of this mutation?

- a. Acts as a serine/threonine kinase to terminate cell proliferation.
- b. Binds adenylate cyclase to overstimulate cAMP production.
- c. Catalyzes the breakdown of PIP_2 .
- d. Is found in the nucleus bound to transcription factors.
- e. Overstimulates the MAP kinase cascade causing abnormal growth.

Other suggested resources:

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

- Scholar Rx Bricks
 - Cell Signaling: <https://exchange.scholarrx.com/brick/cell-signaling>
 - Homeostatic Mechanisms: <https://exchange.scholarrx.com/brick/homeostatic-mechanisms>
- Alberts et al. *Essential Cell Biology*, 6th edition (2023), Chapter 16, “Cell Signaling” pp. 553-593 (hard copy in the library)
- Pollard and Earnshaw *Cell Biology*, Elsevier; 4th edition (2023), Section VII: Chapters 25-28, “Signaling Mechanisms” pp.429-506 .
<https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20180054031>
- Lippincott's Illustrated Reviews: Cell and Molecular Biology, Wolters Kluwer, 3e (2023), Unit IV: Chapters 17-19, “Cell Signaling” pp.179-206.
<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3244>

Lecture Feedback Form:

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>